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AI-TEAM

Software Architecture by Agent Responsibility

Conway's Law Applied to Agentic Software Engineering

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24 March 2026

Department I / Institute of Software (IFS)

WO WISSEN WIRKT.

Agentic Workflows

How coding agents work

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Software Crafters

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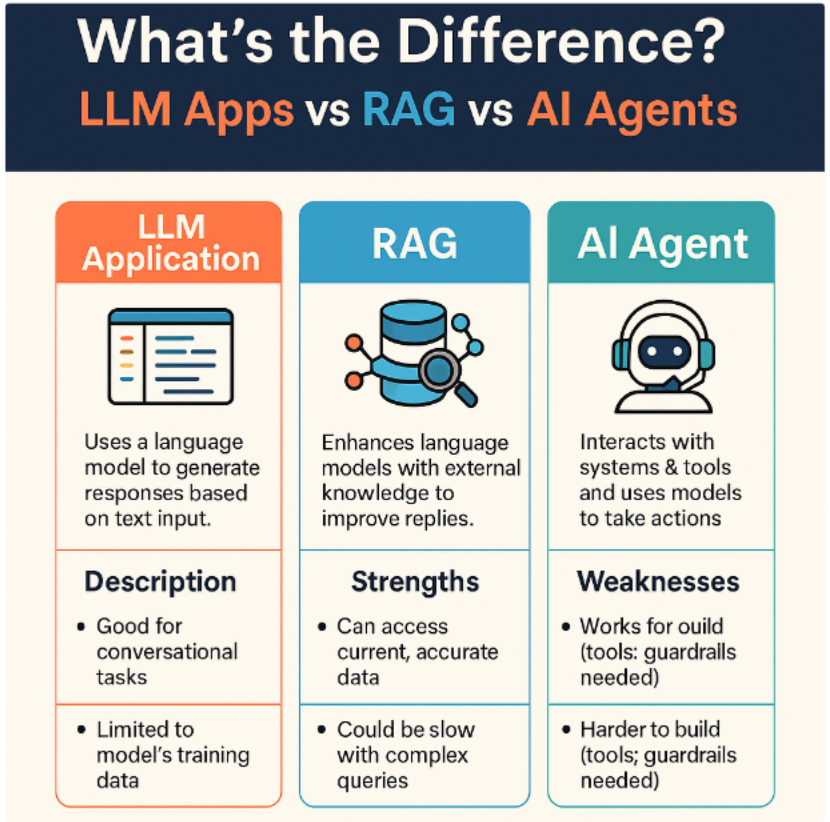
LLMs vs RAG vs Agents



Source: Muhammad Usama Khan (Medium), usamakhaninsights.medium.com

Examples of today's coding agents

GitHub Copilot

Claude Code

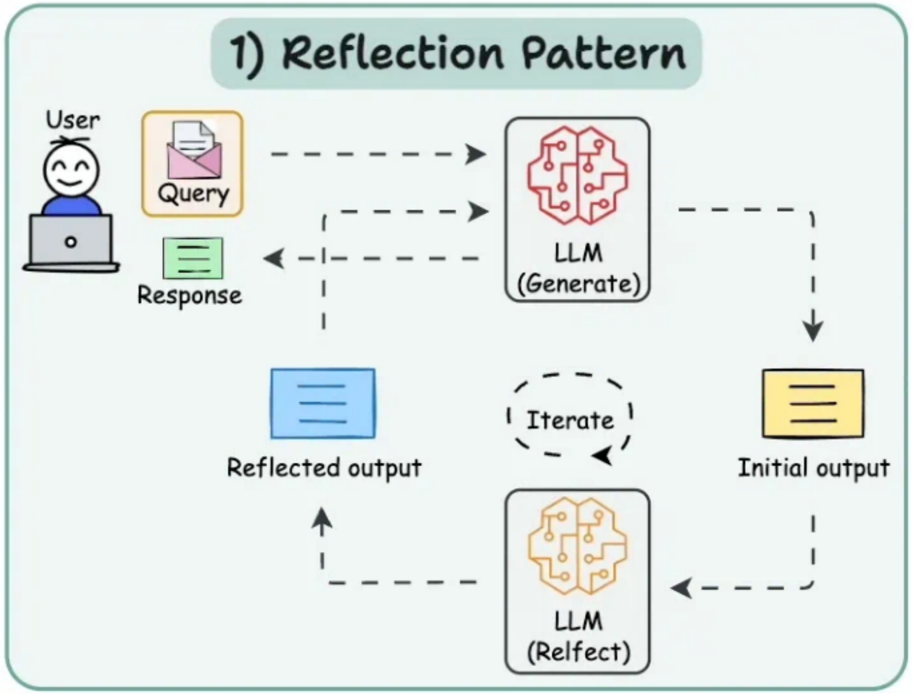
Cline (Open Source)

OpenCode (Open Source)

Codex

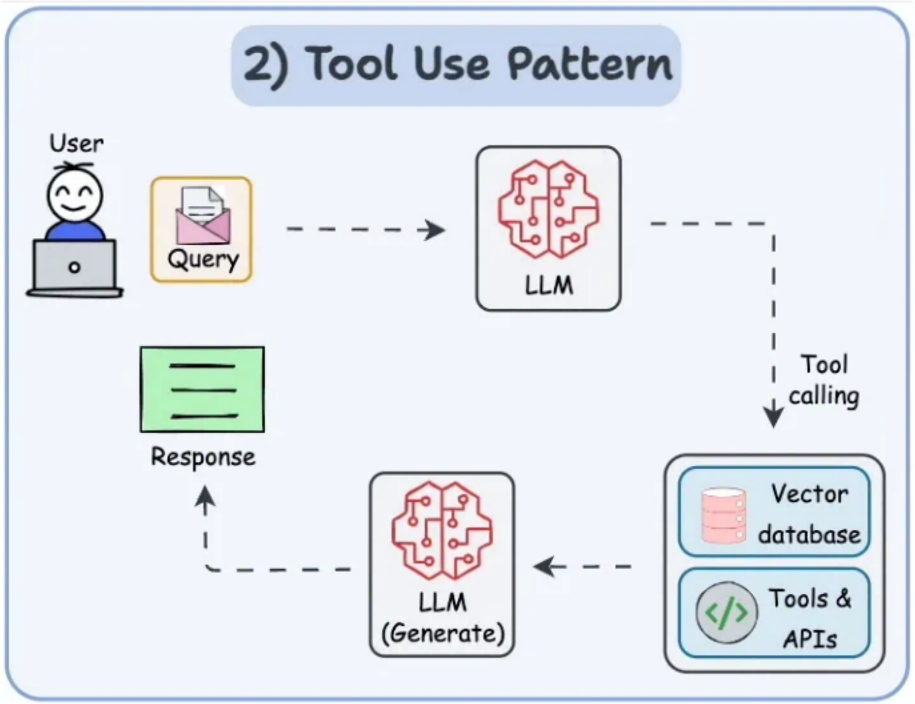
Cursor

Reflection Pattern



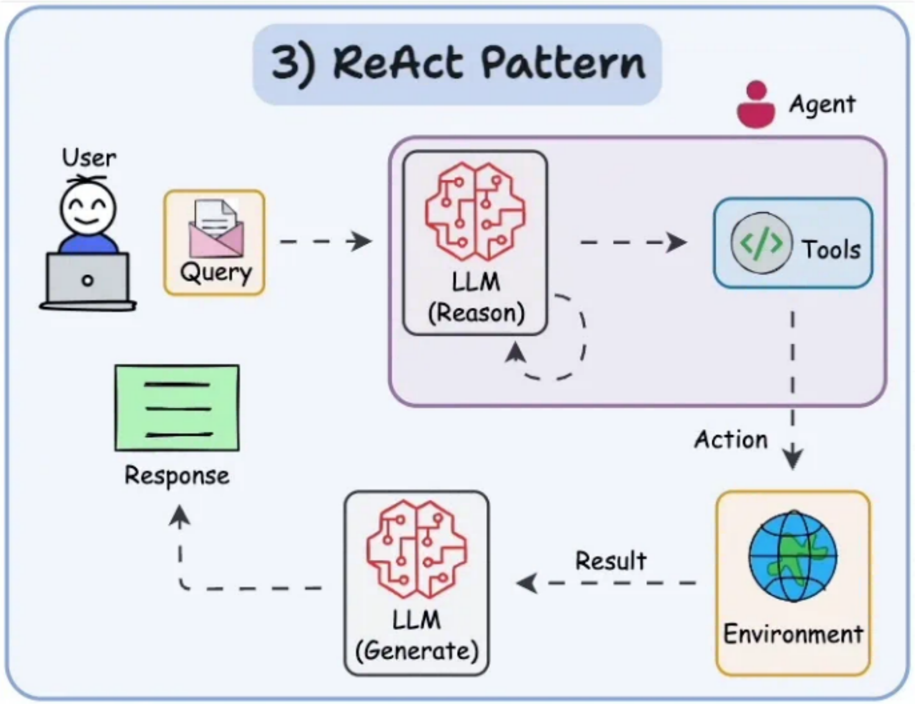
Source: *blog.dailydoseofds.com*

Tool Use Pattern



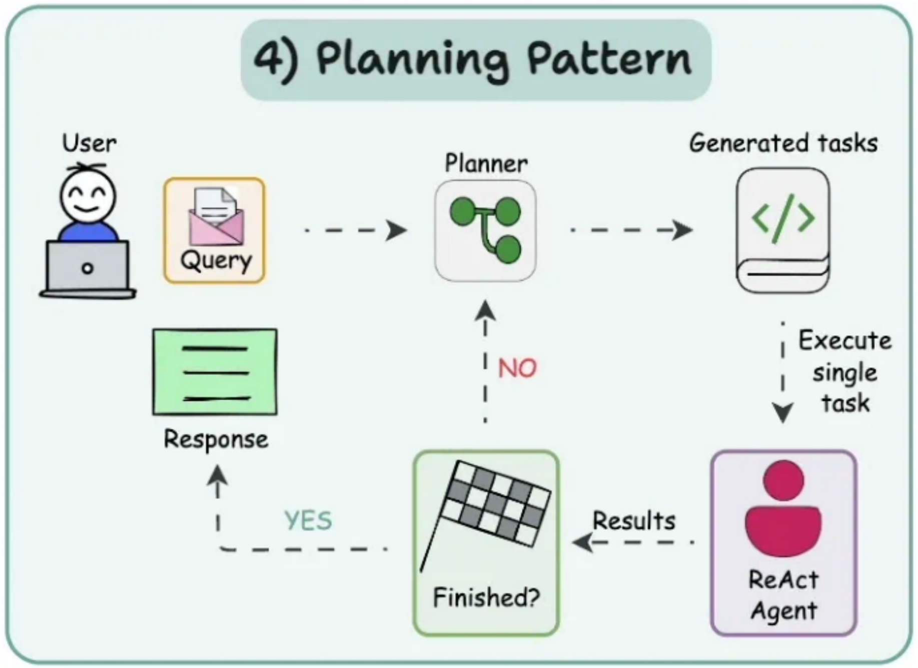
Source: blog.dailydoseofds.com

Reason and Act (ReAct)



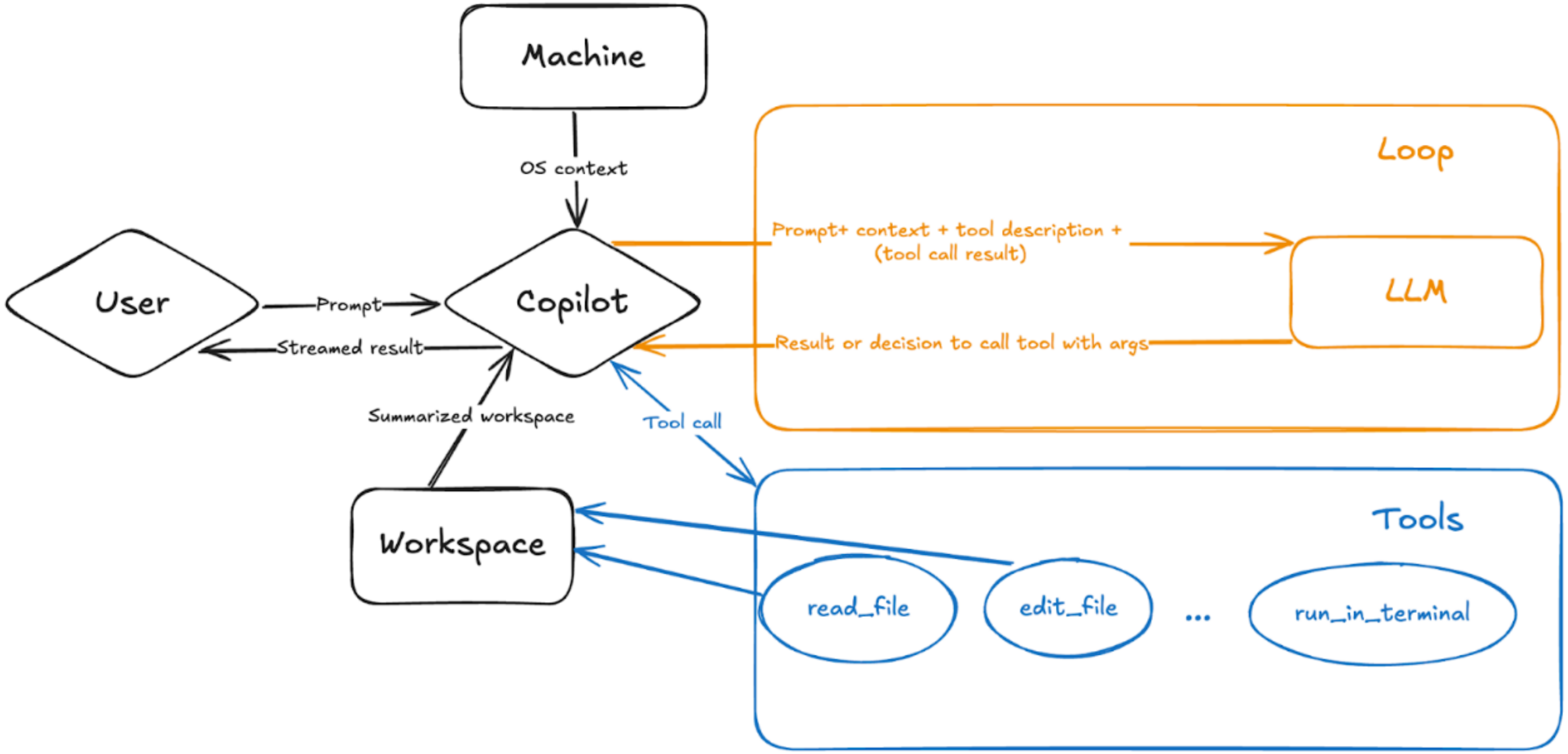
Source: blog.dailydoseofds.com

Planning Pattern



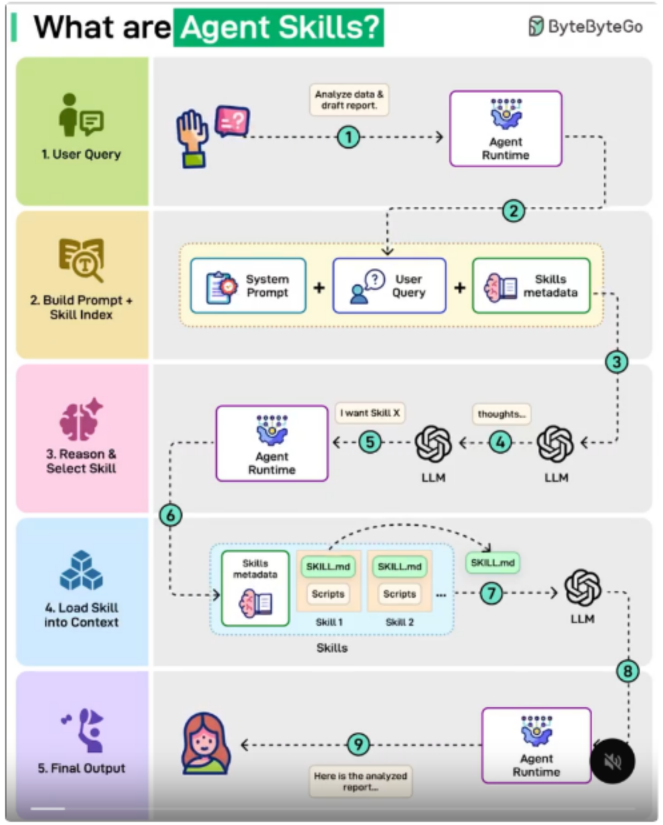
Source: blog.dailydoseofds.com

Agent in Action



Source: VS Code Blog: Introducing GitHub Copilot agent mode (preview)

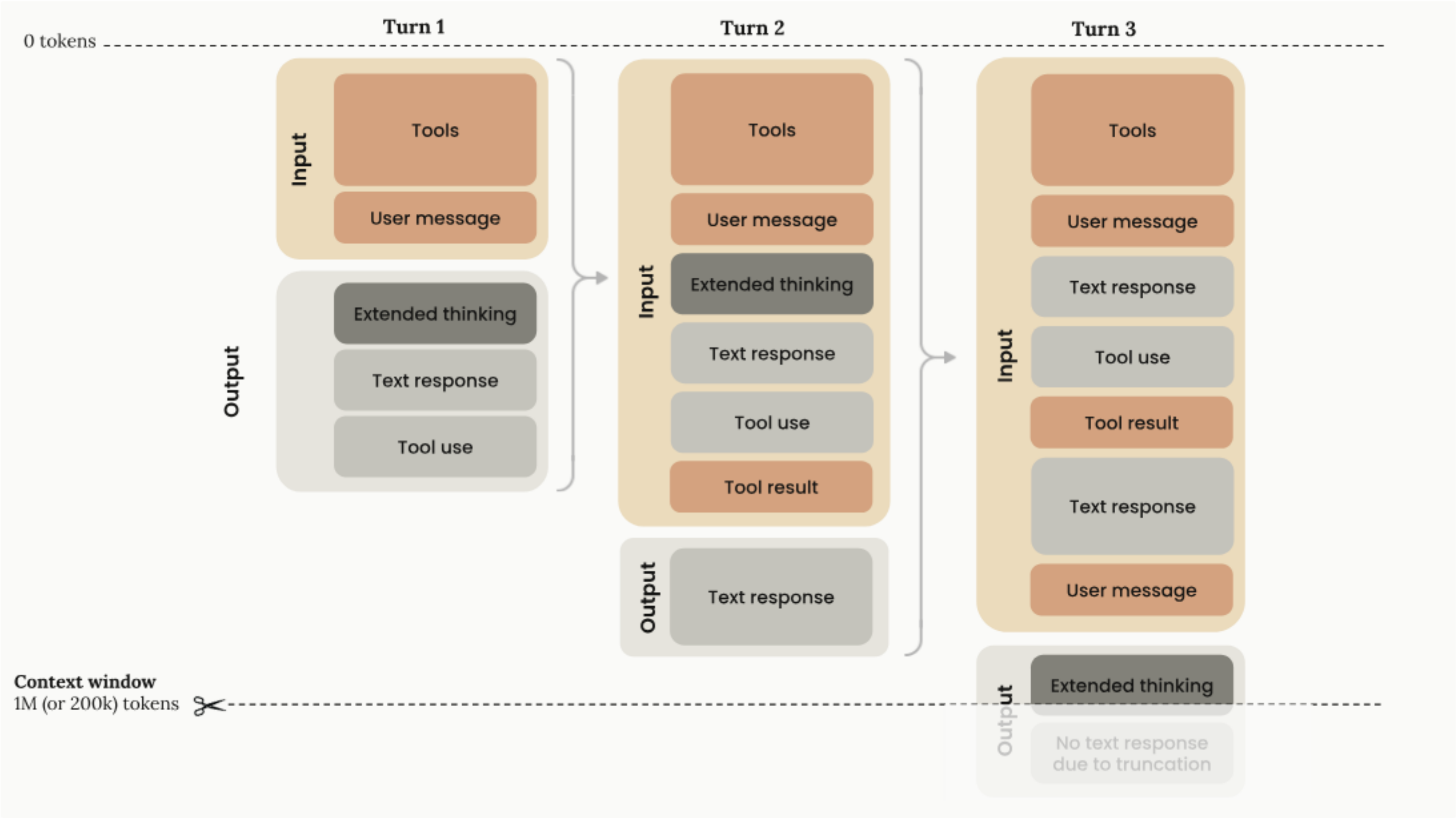
Skills



Source: ByteByteGo, visual about Skills / MCP (ByteByteGo branding is visible in the image).

Agentic Workflows

Context Window



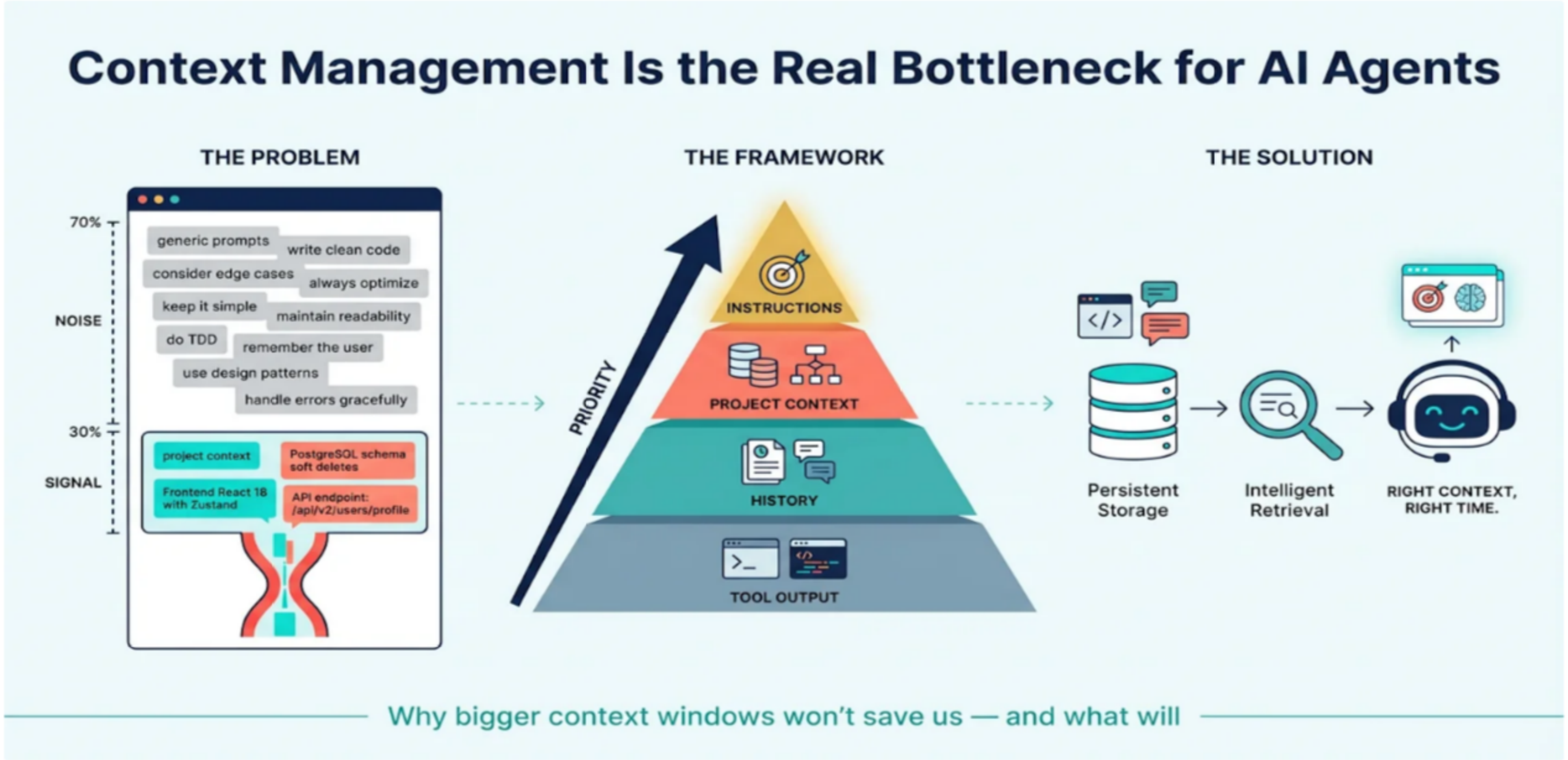
Source: Undetectable AI https://undetectable.ai/blog/claude-vs-gpt-4/

Larger context window = better?

OVERLOAD

Too much context overwhelms the model – focus is lost and response time increases.

Context Window

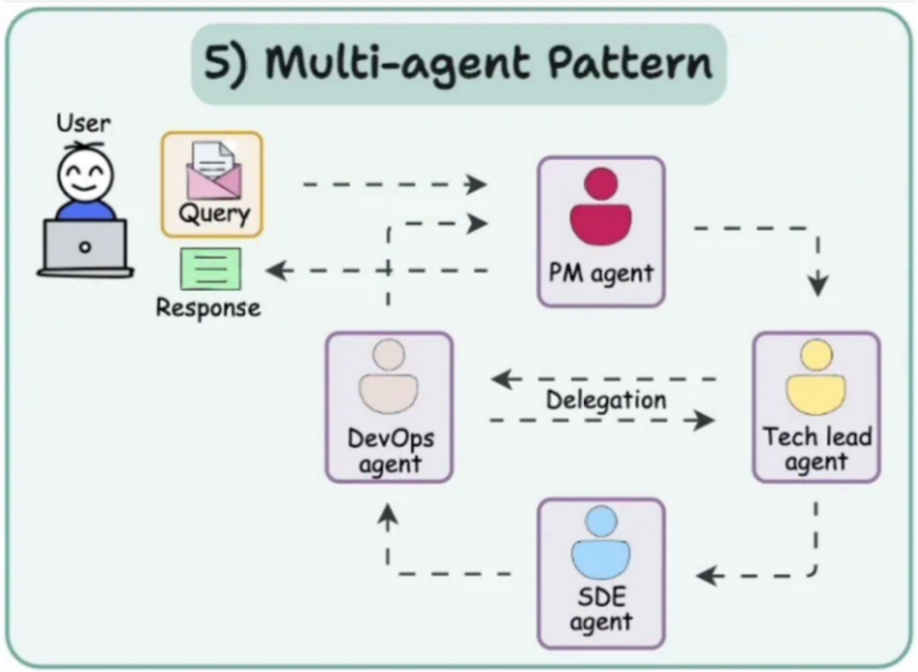


Source: Auxten Wang, Context Management Is the Real Bottleneck for AI Agents (CC BY-NC 4.0).

The problem

- One agent → one context window
- The entire project in a single context window? Opus can do everything! ;)
- Subagents?

Multi-Agent Pattern?



Source: blog.dailydoseofds.com

Conway's Law – Applied to AI

Organization shapes architecture

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Conway's Law

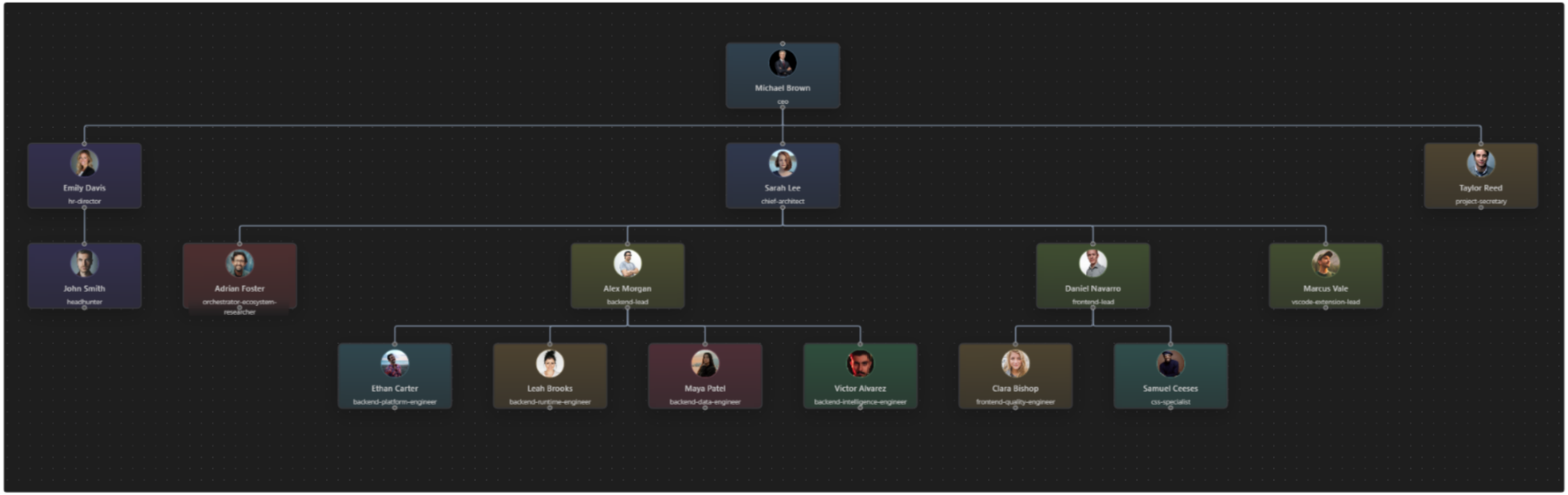
Systems reflect the communication structure of their organization.

- clear structure → clear software
- chaotic structure → chaotic code

🤔

Does this also apply to coding agents?

AI-Team organizational chart



The organization of the AI-Team (a virtual team working on AI-Team!).

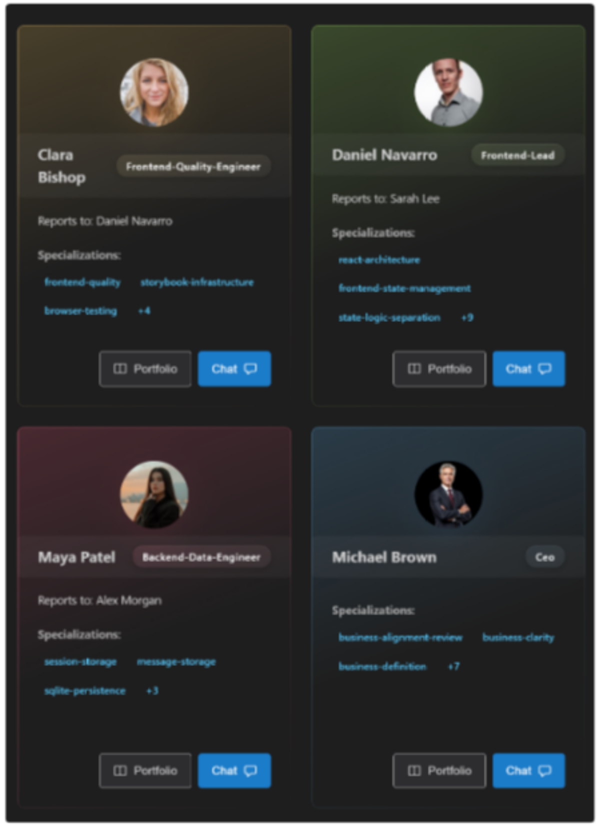
Clear responsibilities in organizations

- focus instead of overload
- clear responsibilities
- better communication
- higher quality

AI-Team creates a suitable agent structure

- clear boundaries in the virtual team
- clear boundaries in the code
- each agent knows its scope

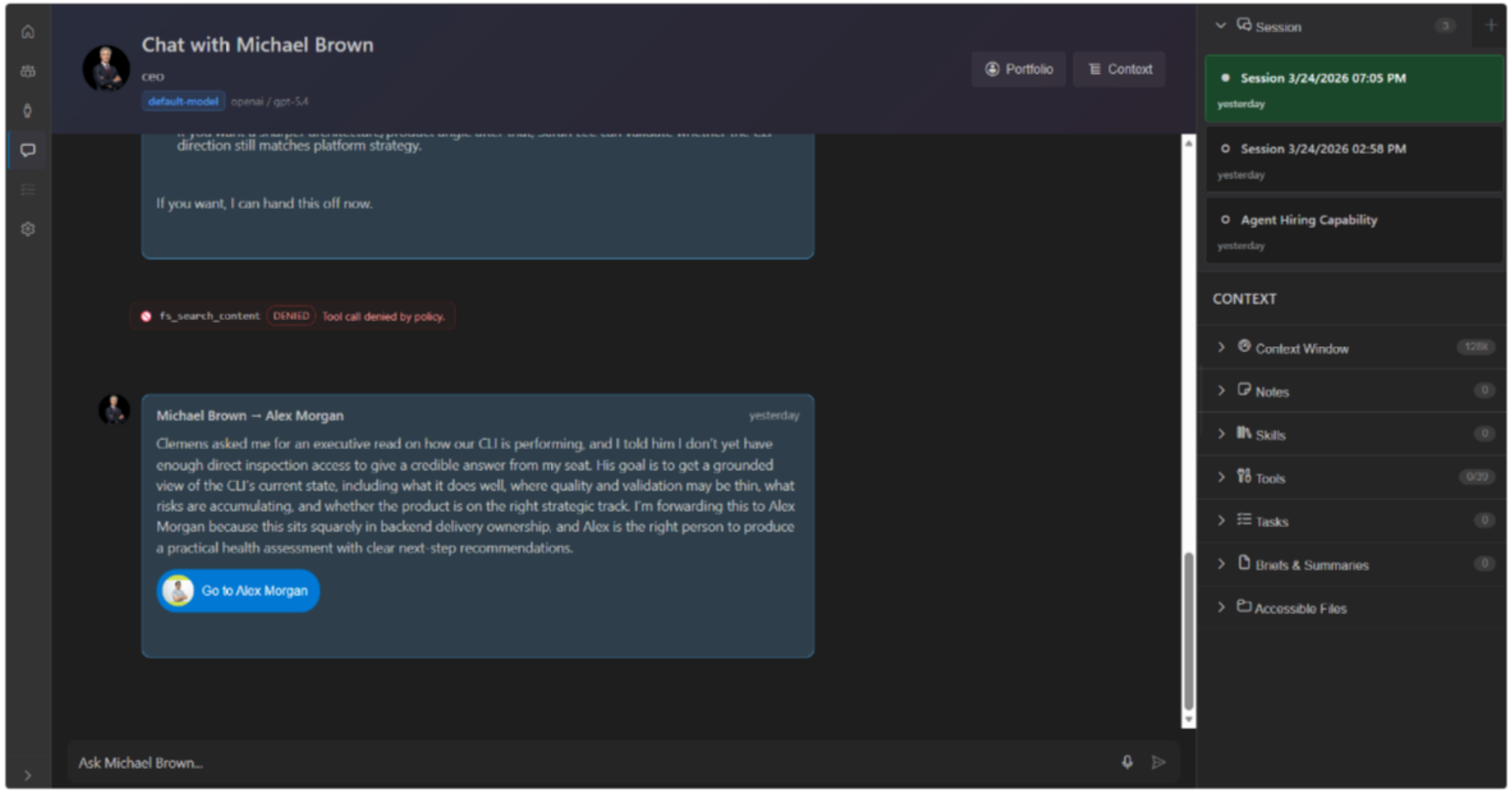
AI-Team Team



Clara Bishop
Frontend-Quality-Engineer
Reports to: Daniel Navarro
Specializations:
frontend-quality
storybook-infrastructure
browser-testing
+4
Portfolio
Chat
Daniel Navarro
Frontend-Lead
Reports to: Sarah Lee
Specializations:
react-architecture
frontend-state-management
state-logic-separation
+9
Portfolio
Chat
Maya Patel
Backend-Data-Engineer
Reports to: Alex Morgan
Specializations:
session-storage
message-storage
sqlite-persistence
+3
Portfolio
Chat
Michael Brown
Ceo
Specializations:
business-alignment-review
business-clarity
business-definition
+7
Portfolio
Chat

Conway's Law – Applied to AI

AI-Team Chat



Change and optimization

- priorities change
- successful teams continuously optimize their structure

AI-Team optimizes the agent team

- calculates responsibilities
- measures the effectiveness of agents
- suggests better structures
- creates new agents when needed

Onboarding & Pair Programming

- New developers (the human kind) need orientation.
- Who knows what area best?
- Which tools fit my task?

AI-Team as navigator

- CSS specialists automatically get Playwright & Storybook
- Backend developers get DB tools & test runners
- A suitable pair programmer is assigned automatically
- Developers find their way immediately because they are automatically connected with the specialists.

Setup is tedious

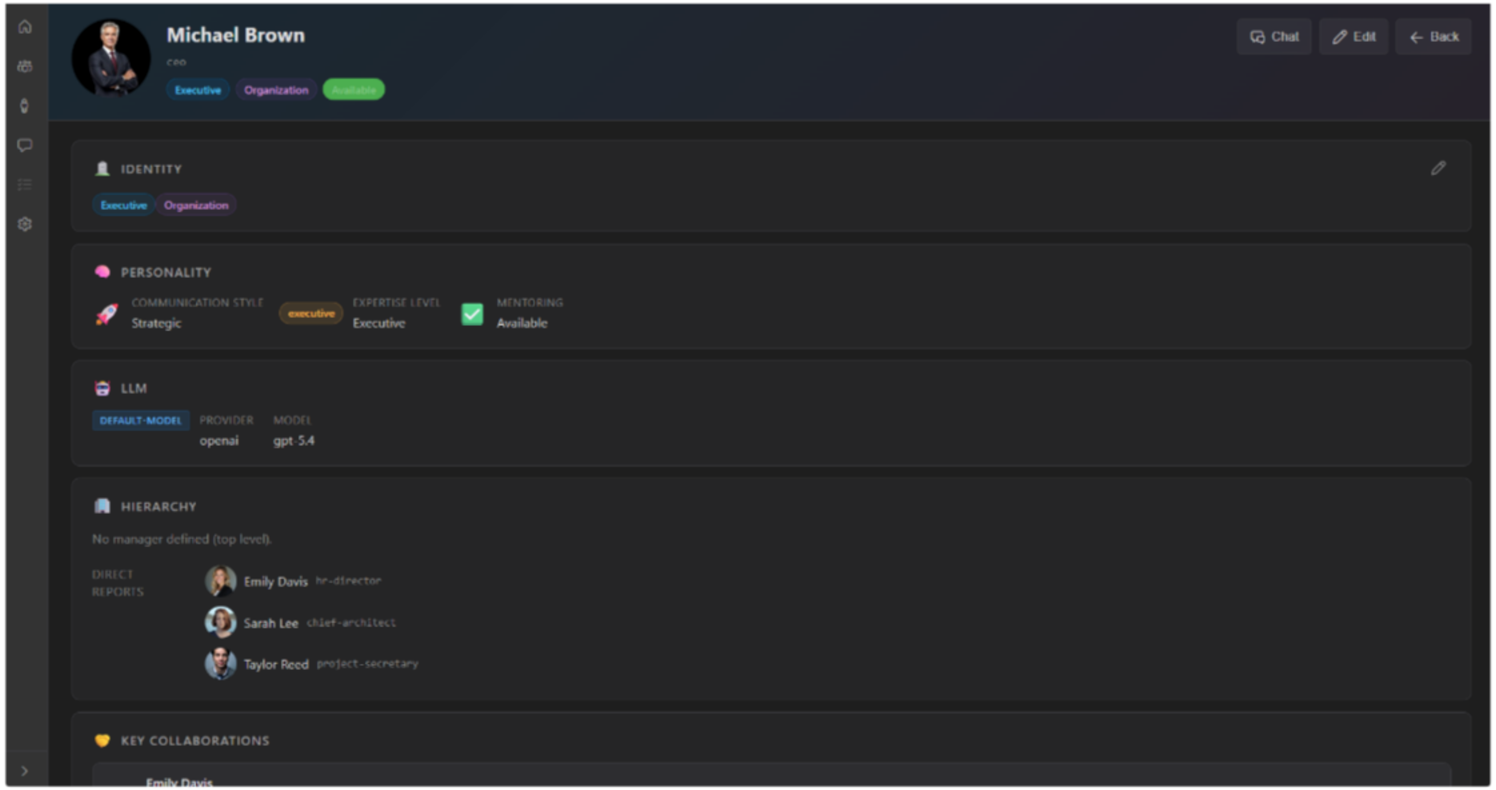
- Prompts
- Tools
- Agent.md, Skill.md, instruction.md
- MCP
- Models

AI-Team makes it human-friendly

- „hire react specialist“
- „hire db expert“
- „hire backend lead“

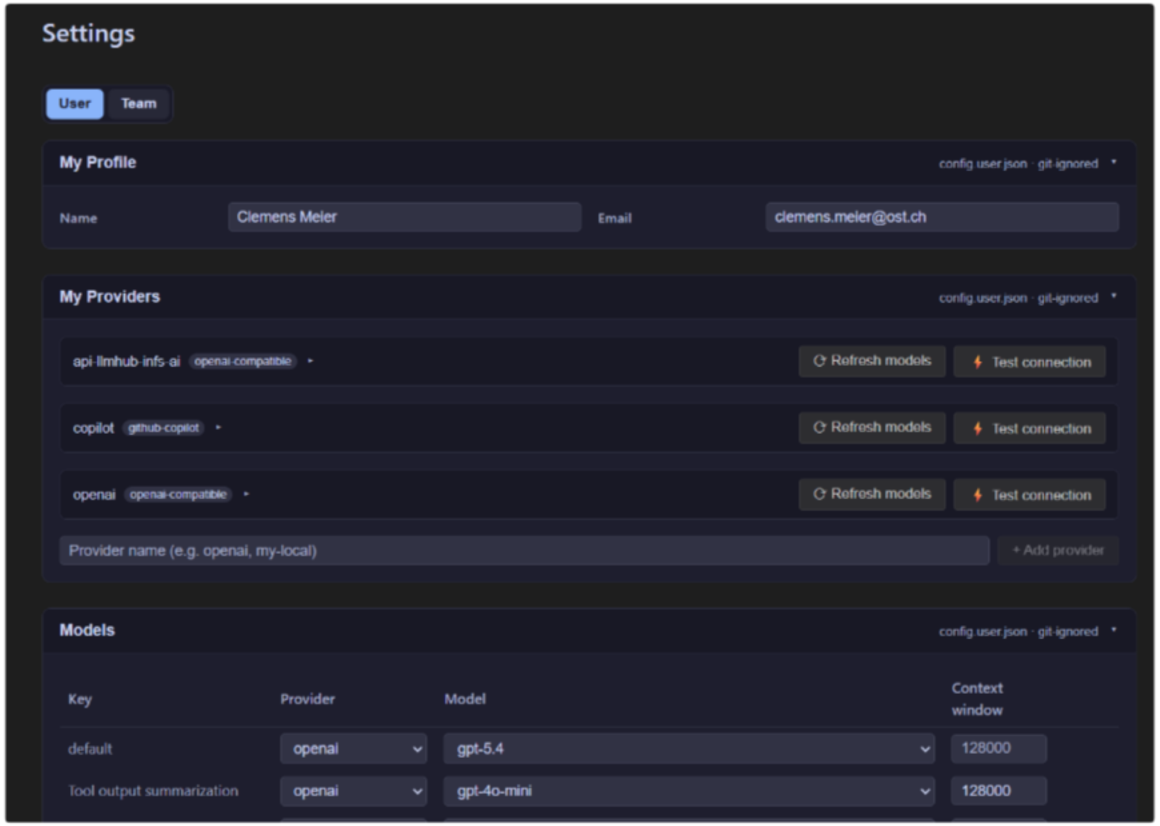
Conway's Law – Applied to AI

AI-Team Portfolio



Conway's Law – Applied to AI

AI-Team Settings



Copilot or Claude Code are data-hungry

- upload huge amounts of code to the cloud
- charge a lot of money for tokens
- create vendor lock-in through contracts

AI-Team:

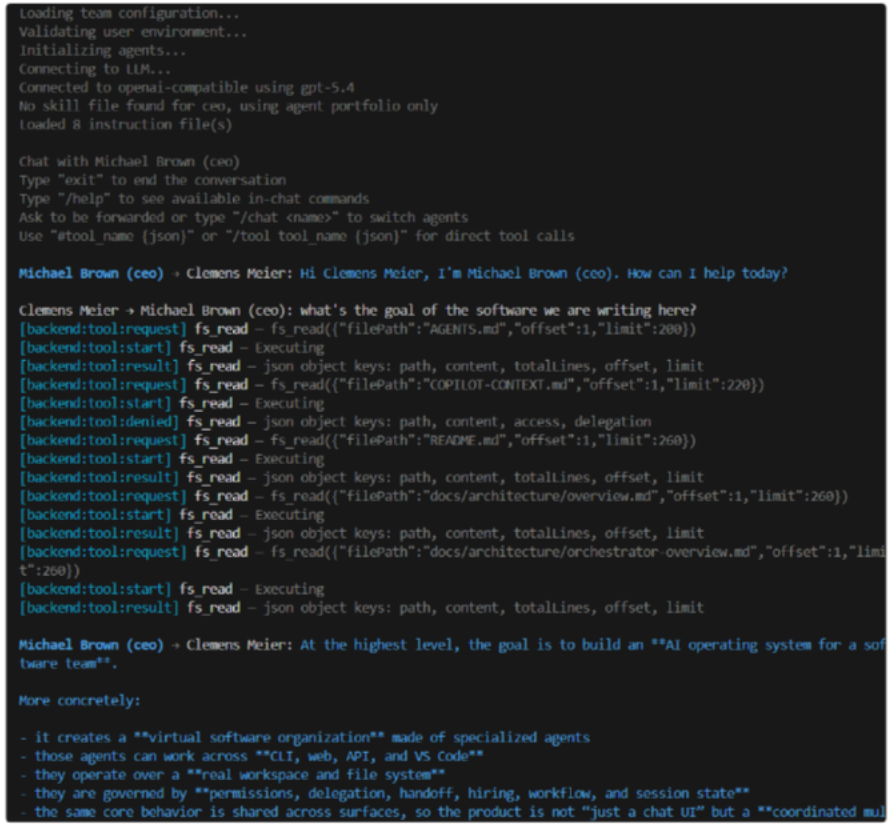
- Open Source
- Bring Your Own Model
- Bring Your Own Provider
- Existing subscriptions can be used

Thesis

- today's agents are powerful – but poorly organized
- Context pollution does not decrease if you give the LLM 100 agents and 1000 skills
- Context cleanup is only necessary when context was not clean from the start.
- We need to bring structure to the agent landscape
- Structure creates quality
- Well-organized agents lead to good software architecture
- Conway's Law also applies to AI

Conway's Law – Applied to AI

AI-Team CLI



Links

Code:

- AI Gitlab-OST Repository (OST internal)
  Attention! Heavy development and vibe-coding chaos ;) Pull requests are not accepted yet, but ideas are very welcome!

Workshops about Copilot:

- Copilot Dev Days Online Workshop
  Highly recommended for anyone who wants to dive deeper into how Copilot works. It offers many practical exercises and examples to help you make the most of Copilot's capabilities.
  A GitHub Academic License (free for OST students with a scanned student ID) is required to complete all exercises, but it is definitely worth it!

- Next Level GitHub Copilot (interesting website)